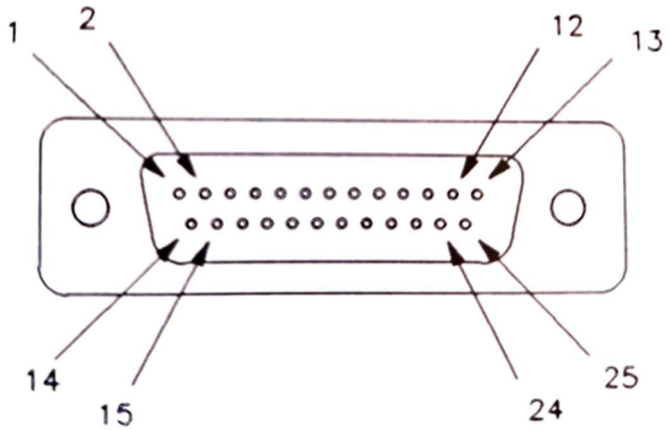
Appendix F

Interfacing

F.1 25-way D type (COM1)

This socket is for connecting to the main serial port, COM1. It may also be used to connect to the auxiliary serial port (COM2), as all its pins are duplicated on this connector.



Pin	Description	Pin	Description
1	Protective Ground	14	TXD1 (COM2)
2	Transmitted Data	15	Transmit Clock
3	Received Data	16	RXD1 (COM2)
4	Request to Send	17	Receive Clock
5	Clear to Send	18	(Not used)
6	Data Set Ready	19	(Not used)
7	Signal Ground	20	Data Terminal Ready
8	Data Carrier Detect	21	(Not used)
9	(Not used)	22	Ring Indicator
10	+5V Power Out	23	(Not used)
11	Reserved	24	(Not used)
12	(Not used)	25	Charger In
13	CTS1 (COM2)		

The electrical specification of this interface is in accordance with EIA specification RS232C:

Outputs	Logic 0: >= +5V into 3.3 Kohm load
	Logic 1: <= -5V into 3.3 Kohm load
Inputs	Logic 0: +3.5V ... +25V
	Logic 1: +2.5V ... -25V

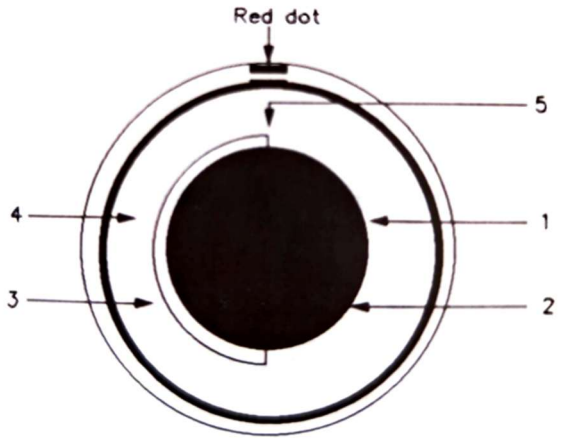
All data input pins present an impedance of 6.8 Kohm.

Pin 10 supplies +5V DC +0.5/-1.0V at up to 100 milliamps, under program control.

F.2 5-way FISCHER (COM2)

This socket is for connecting to COM2, or connecting a barcode wand or the **Hunter 16** AC adaptor.

Note: The only AC adaptor that you should use is the **Hunter 16** AC adaptor with a 5-way FISCHER plug.



Pin	Description
1	Charger in
2	RX Data
3	Clear To Send
4	TX Data
5	Ground

These pins are duplicated on the 25-way D type, for your convenience.

Note that pin 3 (CTS handshake input) is used for barcode wand input. This means that you cannot use COM2 handshaking and barcode input simultaneously, but this is unlikely to cause problems in practice.

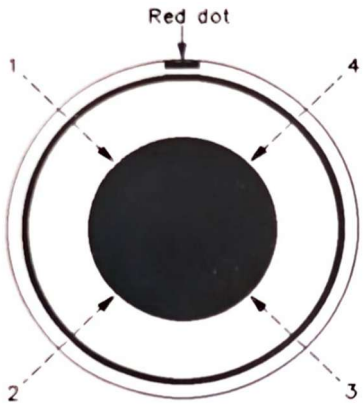
The electrical specification of this interface is in accordance with EIA specification RS232C:

Outputs	Logic 0: >= +5V into 3.3 Kohm load
	Logic 1: <= -5V into 3.3 Kohm load
Inputs	Logic 0: +3.5V ... +25V
	Logic 1: +2.5V ... -25V

All data input pins present an impedance of 6.8 Kohm.

F.3 4-way LEMO

This socket is for connecting a suitable mains adaptor or single bit inputs such as barcode wands, voltage free switch contacts, etc.



Pin	Description
1	Ground
2	Input
3	Mains adapter input
4	+5V out

Pin 2 is configured as a protected, pulled up CMOS compatible line, with a logic 1 threshold of > +3.5V and a logic 0 < +2.5V. It is protected to +25V. The pull-up resistor enables the use of a switch to ground to provide the input.

No debounce circuitry is provided. If it is required then software must be written for this purpose.

Note that pin 2 is duplicated on pin 3 of the 5-way LEMO, where it is used for the COM2 CTS handshake input. This means that you cannot use COM2 handshaking and barcode input simultaneously, but this is unlikely to cause problems in practice.

Pin 4 supplies +5V DC +0.5/-1.0V at up to 100 milliamps, under program control (INT 15H function 81H: see Part 3).